

9. Artificial Intelligence for On site Analysis of Microplastic in Urban Water Runoff

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ABSTRACT

The topic microplastics has become an increasingly challenging subject posing ecological risks and hazards to human health. Current detection methods are expensive, lab-based and time-consuming. This literature survey aims to explore the possibilities of combining current state-of-the-art machine learning approaches into increasing efficacy in microplastic detection. Machine learning models such as YOLO, Faster R-CNN and Edge Detection are promising techniques used in instantaneous object detection. These methods can be used in combination with fluorescent staining of microplastics to potentially quantify, identify and analyse microplastics contained in urban water runoff samples.

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INTRODUCTION

Urban pollution has become an increasingly widespread problem, especially in recent years after the industrial revolution leading to population growth. Various forms of pollution, including air, water, soil, noise, and light, not only harmful to the environment but also degrade quality of life and pose serious health risks. Among the most harmful pollutants are plastics and microplastics.

Ever since the plastic boom between the 1960s and 1970s, the use of plastics has been widespread, and its polymer breakdown has been inevitable. The first signs of plastic pollution in marine life were reported in the Science journal¹ in 1972. Research conducted by Carpenter and Smith highlighted that the extensive use of artificial plastics in products such as facial scrubbers and paint has been a major contributor to the presence of microplastics in the environment.¹ An overview has mentioned the deterioration and weathering effects have led to larger plastic fragment pieces into smaller and smaller pieces. As a result of the Pacific gyres, the clusters of these small pieces present in the marine environment have led to the formation of “The Great Pacific Garbage Patch” - ultimately the fate of unwanted waste plastics.² The article further mentioned that the sources of pollution in the marine environment such as, illegal dumping, winds blowing household waste into the river, treated/untreated sewage to name a few. In a more detailed scale, an article mentioned the adverse effects of microplastics on human health, touching on topics such as the plastic pathways, ecological systems and microplastic contaminations in the human food chain. Furthermore, it provided a detailed listed table of marine species with microplastic.³ The marine species mentioned such as carp fishes⁴ and Atlantic herring⁵ have been reported to contain microplastic in the stomach. Mamun *et al.*³ mentioned Campanale *et al.*⁶ and his works on the toxic effects of consuming heavy metals adsorbed onto small plastic polymer structures. Potential risks of human infertility, gastrointestinal damage, carcinogenic effects were discussed and the paper concluded that microplastics in the human body pose serious health risks and irreversible damages to human life.

The topic of this project is to stain various microplastics obtained from urban water using a fluorescent dye and capture these microplastic pieces under a smartphone camera in certain conditions. Nile Red (IUPAC name: 9-(Diethylamino)-5H-benzo[a]phenoxazin-5-one) has been traditionally used as an inexpensive dye to highlight the microplastics fragments for clearer visualization with the naked eye²⁹. In a comprehensive review paper⁷ on the topic of microplastics and machine learning identification analysis techniques, thorough works have been done previously to stain microplastic fragments using Nile red dye and determine the wavelength that yields the

strongest intensities for various microplastics. The main identification techniques highlighted by the review paper were the use of spectroscopic imaging systems, which are costly and time-consuming for quick identification. In the machine learning algorithm section, the focus was emphasised on identification and analysis of ATR-FTIR spectra. It was hoped to reduce the amount of manual intervention by creation of a semi-automatic analytical method for microplastic characterisation. However, prior to microplastic analysis, a series of time-consuming microplastic extraction methods would need to be performed. Hence, the topic of this literature review aims to explore a range of literature in the areas of: Microplastics in urban water, the structures of these polymers, the use of fluorescent dyes to 'tag' these microplastics, the existing quantification approaches of microplastics and investigate whether machine learning can be used to identify microplastics based on the absorption intensity of the fluorescent microplastic particles.

PLASTICS, MICROPLASTICS & POLYMERS

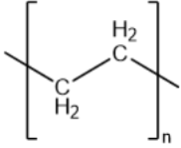
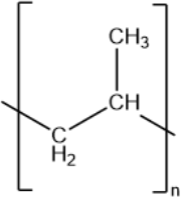
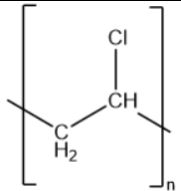
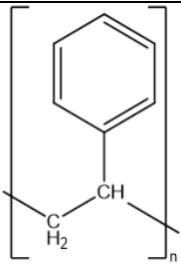
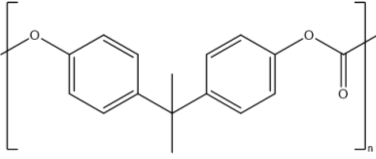
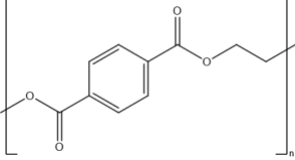
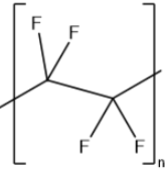
Plastics gained their popularity due to their inexpensive synthesis while being versatile in its chemical and physical properties, such as flexibility, mouldability, durability and chemical resistance. The original outlook of a creation to substitute the scarcity for ivory and tortoiseshells in the mid-1800s, led to the invention of 'Parkesine' by Alexander Parkes⁸ eventually leading to 'Celluloid', one of the first plastics. Later in the early 1900s, Belgian chemist Leo Baekeland patented 'Bakelite', a phenol and formaldehyde resin structure, marking the early footsteps of plastic invention⁹. A failed experiment combining ethylene and benzaldehyde in the mid 1900s in the Imperial Chemical Industries in England, led to the invention of polyethylene¹⁰ and marked the beginning of plastic pollution. In the Textbook of Polymer Science published by Wiley¹¹, plastics are defined as synthetic organic polymers of repeating molecular units, monomers. These synthetic polymers are often formed through polymerisation processes. Table 1 in the Appendix section, shows some commonly used plastics, their chemical structures and properties.

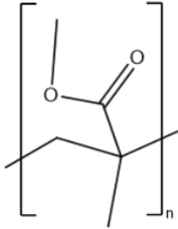
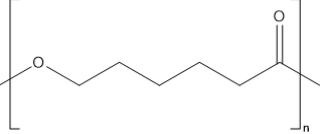
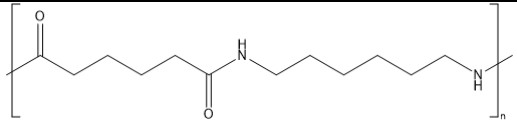
The papers^{7,12} mentioned the sources of microplastics and categorized them into primary and secondary source microplastics. Primary sources of microplastics are usually synthesized in the manufacturing process, whilst secondary sources of microplastics are formed as a result of plastic deterioration. Several studies^{13,14,15} explored the manufacturing of microbeads used in facial scrubbers, toothpaste, shower gel. Extraction and analysis methods used were dissolving products

in boiling water, then vacuum filtration, oven dried the sample microbeads, FTIR and Raman spectroscopy analysis on the material, quantification using Image J and a detailed image of the microbeads were captured. These microbeads were found to be of polyethylene. Cheung and Fok¹³ aimed to estimate the numerical emission of these microbeads into the seas near China, whilst Napper *et al.*¹⁴ aimed to quantify the single use emission of these microbeads in the UK. Their papers concluded that in one single use of facial scrubbers approximately less than 100,000 microbeads are released in the sewage system. The estimated quantitative numbers are important figures, as it will give a perspective of the amount of microbeads that may be present in the urban runoff water samples.

While microbeads are a common source of microplastics, synthetic textiles also contribute heavily to the sources of plastic pollution. Fibre fragments or microfibrils are released during the textile manufacturing process or during household washing⁷. An insightful report by Manshoven and Christis¹⁶ provided an overview on the current understandings on the release of fibre materials from textile industries in Europe. Similarly, Salvador Cesa *et al.*¹⁷ also outlined that the most commonly found microplastics are fibre fragments of types: polyester, acrylics, nylon and polypropylene in wastewater effluent. Current understandings suggest that during household washing, fibre fragments are released due to mechanical stress and collision within the washing machine. The fibre fragments then follow the wastewater disposal pipes into the sewage system. Furthermore, in the article by Napper and Thompson¹⁸ experimental household washings were performed on 6 kg textile clothing wash under certain conditions. The fibre fragments were collected at the wastewater outlet with a sieve of pores sized 25 µm. For the first 6 kg polyester wash, 2.79 mg of polyester fibre fragments were released. The report¹⁶ mentioned the efficacy of microplastics removal in the process of wastewater treatment with multiple stages: physical, chemical separation, biological treatment and oxidation processes. Despite efforts to retain microplastics and microfibrils in wastewater treatments, untreated wastewater and sludge still contain immense amounts of synthetic polymers. The fate of these synthetic polymers is the release into the environments, agricultural soil and eventually rivers, seas and oceans. The report is insightful because it emphasised the possibility of fibre strands contaminants in the urban water runoff samples that will be collected.

Table 1. Tabular information of some commonly used plastics, with chemical structures, uses and their polarities. Structures were drawn using ChemDraw.

Name of Polymer plastic (Abbr.)	Chemical structure (Repeat unit)	Some Uses	Chemical Properties
High density polyethylene (HDPE) Low density polyethylene (LDPE)		Insulators, pipes, toys	Non-polar
Polypropylene (PP)		Water pipes, integral hinges	Non-polar
Poly vinyl chloride (PVC)		Adhesives, records	Polar
Polystyrene (PS)		Light panels, lenses	Non-polar
Polycarbonates (PC)		CDs, protective gear, medical equipment	Polar
Polyethylene Terephthalate (PET)		Bottles	Polar
Polytetrafluoroethylene (PTFE)		Nonstick surfaces, gaskets	Polar

Polymethylmethacrylate (PMMA)		Bathroom fixtures, knobs, illuminated signs	Polar
Polycaprolactone (PCL)		Bone repair, film and foam plastic	Polar
Nylon 66		Clothing, textiles	Polar

Degradation of Polymers

A critical review article¹⁹ demonstrated the plausible mechanisms of the degradation of plastic polymers in marine environment into microplastics. The mechanism for pure polymer degradation though simple, under real world conditions a degree of complexity is present. The challenges lie on various chemical additives used to stabilize plastic polymeric structures.²⁰ The enhanced structure stability of these polymers allows a longer plastic longevity, but in exchange, these waste plastics take a significantly longer time to degrade. In another critical review article²¹ focused on the degradation of synthetic textile polymers, categorised into the pathway processes ‘abiotic degradation’ and ‘biotic degradation’. Photodegradation of polymers is heavily discussed in Gewert *et al.*¹⁹ as it is the most plausible process that arises in marine environments. The article mentions that photodegradation heavily relies on oxygen and UV-radiation, which allow radical mechanisms to occur between carbon-carbon structures. This is the case with polyethylene, polypropylene, polystyrene and PVC plastics which are heavily exposed to sunlight and oxygen whilst in the oceans. The main mechanism is introduced in 3 stages: initiation, propagation and termination which leads to chain scission, thus breaking large polymer into smaller polymers, hence microplastics. The initiation stage focuses on UV-radiation from sunlight attacking the carbon-carbon backbone structure, forming carbon radicals. The carbon-hydrogen bond dissociation energy is 413 kJ/mol. In the propagation step, these carbon radicals attack oxygen present in the air and form peroxy radicals and hydroperoxides²². Chain scission occurs in the termination stage where, the shorter polymer

chains contain olefins, aldehydes and ketone groups as a result of repeated oxidation processes. The carbonyl groups present has enhanced the hydrophilicity in the shortened polymer chain. This allowed the involvement of biotic degradation, as microorganisms further decompose short polymer structures into monomers. Understanding the mechanism of polymer degradation in marine environments is relevant to this research, as it provides an explanation for the formation of microplastics. As mentioned above, short polymer chains found in marine environments are most likely to contain polar carbonyl groups, the use of Nile red may not be as effective to stain these polar compounds. In further extent on the topic of this literature survey, the use of microorganisms to fully decompose microplastics present in the environment is reasonable and a practical approach in the near future.

The article by Cai *et al.*²³ in comparison to the article by Gewert *et al.*¹⁹, focused on the biotic degradation of plastics. Both articles highlight a common understanding: the lack of functional groups and hydrolysable bonds in plastic polymers makes it difficult for microorganisms to attach, which is why the biotic degradation of plastics remain a challenge without previous abiotic degradation. The article²³ listed a summary of potential microorganisms such as bacteria, fungi and enzymes with their corresponding plastic carbon food sources and optimum temperatures for degradation. For example, the PET plastic structure contains hydrolysable carbonyl bonds which are suitable for microorganisms to attach to. Both articles^{23,24} mentioned the potential use of the bacteria *Ideonella sakaiensis* 201-F in biotic degradation processes. However, the optimum temperatures for this degradation process are between 70-75°C. Though these processes are feasible biotic degradation for short plastic polymers, in urban water situations perhaps is unfeasible.

In real world applications, biotic degradation of plastic polymers does occur alongside abiotic degradation situations. In the context of microplastics in urban water runoffs, pure biotic degradation without the aid of abiotic degradation, seems unlikely. Abiotic degradation of radical and oxidation reactions though highly likely in marine environments, in sewage wastewaters constant UV radiation exposure does not appear to be plausible. As a result, the microplastic fragments found in urban water runoff are likely to be a result of primary sourced microplastics or microplastics resulting from prolonged physical degradation due to mechanical stress.

MICROPLASTICS IDENTIFICATION & QUANTIFICATION

Traditionally, microplastics have been identified through spectroscopic approaches²⁵. The techniques used for quantitative and qualitative analysis include the use of: Fourier transform infrared (FTIR) spectroscopy²⁶, Raman spectroscopy²⁷, scanning electron spectroscopy, pyrolysis gas chromatography coupled with mass spectroscopy (Pyr-GC/MS)²⁶, and liquid chromatography (LC). In Table 3 in the study done by Li *et al.*²⁵, a comparative table of the various methods used to identify microplastics and their respective advantages and limitations were listed. ATR-FTIR is a common approach used for microplastic analysis due to the characteristic peaks it produces on the spectra. This allows for easy recognition and identification of the specific plastic polymer types that are present in the water samples. However, some limitations include the high cost of the instrument and its lack of portability, restricting most of the analysis to laboratory environments. One of the major drawbacks of FTIR is the need for dry microplastic samples, which encompasses the need for isolation and extraction techniques prior to microplastic analysis. Hence, despite the emergence of automatic FTIR imaging, the use of FTIR for microplastic identification, while highly accurate, remains time-consuming²⁶. A similar limitation is also present in Raman spectroscopy. Despite an effective imaging system; coherent anti-stokes Raman scattering (CARS) microscopy²⁷ has been developed to visualise wet samples of microplastics of sizes 2-3 μm , this technique remains time-consuming and expensive. In the case of Pyr-GC/MS, it seems only feasible in laboratory environments due to the pyrolysis of samples at temperatures of 700°C²⁶. Lacking in portability, it does not seem a practical approach for microplastics identification.

Quantification Approaches

The aim of this project is to build a novel portable black box which uses fluorescence light and smartphone-based imaging system to stain and visualise microplastics present in urban water runoffs. The use of artificial intelligence then enables rapid identification and quantification of the microplastics from these water samples. A study²⁸ similar to this project has been performed, which uses AI object detection techniques to live track and detect microplastics in underwater environments, focusing on low power applications. Their methodology is physics and computing based, and the velocity of the flowing water has been taken into account. Using YOLOv5 and DeepSORT, they successfully detected and quantified microplastics in waters of up to 34 cm/sec velocity, provided that the microplastics were 5 mm in size. At water velocities of over 34 cm/sec,

microplastics were not detected. These results were binary outcomes, i.e. successful detection or no detection, hence if a microplastic was detected it will be quantified. The live tracking imaging device used by the study was a Logitech C270 HD webcam with a maximum video resolution of 720p at 30 FPS (frames per second). Given that their approach of using an imaging device of 720p gave promising results, it is reasonable to infer that using a smartphone camera of higher resolution and quality, under enclosed environmental conditions will also yield promising microplastic quantification results. This article provided valuable information for this literature survey as it confirmed that using YOLO object detection models for live microplastic detection is achievable, especially at high flow rates. Furthermore, it highlighted the existing knowledge gap of combining a more chemistry-based approach for low-cost, rapid microplastic identification. In comparison to the article above²⁸, the study done by Prata *et al.*²⁹ used Nile red fluorescence tagging in combination with a script-based automated quantification techniques, “Microplastics Visual Analysis Tool (MP-VAT)” which was integrated with ImageJ. Their results were valuable as it provided a table with the fluorescence intensities of different types of plastics excited at different wavelengths. Their use of MP-VAT as a microplastic quantification approach appears time-consuming, especially in using a combination of various applications such as Microsoft Excel and Adobe. The use of a machine learning object detection model would reduce the amount of time needed to quantify microplastics. These two research articles offered valuable insights and showed promising results, contributing to this literature review. The project appears to combine approaches from both studies and will yield positive outcomes.

Nile Red and its Derivatives

The topic of this project is to stain microplastics present in urban water runoff samples with a fluorescence dye. Fluorescence dyes are often used as tagging methods²⁹, to visually highlight desired compounds. The study done by Prata *et al.*²⁹ discussed the necessity of staining microplastics as a means of differentiating between microplastics and other mineral and organic materials. The stronger the fluorescence intensity, the easier it is for these fluorescent microplastics to be photographed, especially using a smartphone camera, described in the context of this project.

After reviewing a series of literatures, some possible fluorescence dye options are: Meldola’s blue, Nile red²⁹, Nile blue, Rhodamine B dye³⁰, Rose Bengal³¹. In the article by Jose and Burgess³², they categorised fluorescence dyes into two categories, phenoxazines and benzophenoxazines. The article further stated, dyes need a strong electron withdrawing or donating substituent for good

absorbance characteristics and fluorescence properties. The chemical structure of microplastics are repetitive carbon-carbon polymer structures, such as low-density polyethylene (LDPE), high density polyethylene (HDPE), polystyrene (PS), poly vinyl chloride (PVC). A mixture of polar and non-polar polymer structures is contained within microplastic samples. Obtained urban water runoff samples are likely to contain a mixture of synthetic fibres, microplastics and contaminants such as chitin, wood or chalk. After having compared the specificities and chemical configuration structures of the fluorescence dyes, Nile red and Nile blue seemed the most applicable fluorescence dye to use to stain microplastic structures. The major advantage of Nile red is the inexpensive cost and the high fluorescence properties due to the 9-diethylamino substituent; the electron delocalisation of the structure leads the carbonyl group as the final electron acceptor. Another suitable fluorescence dye is Nile blue³³, the molecule is positively charged compared to Nile red, a neutral molecule. Similar to Nile red, Nile blue has a 9-diethylamino substituent which readily delocalises the rings with iminium substituent as the final electron acceptor. Absorption and emission spectra of these two fluorescent dyes are also within the same range, varied between different solvents³⁴. A limitation of Nile blue compared to Nile red would be the short fluorescence lifespan, Nile blue measured 1.42 nanosecond whilst Nile red measured 3.65 nanosecond³². Studies by Grofcsik *et al.*³⁵ show that the fluorescence lifespan is affected more by temperature and polarity of solvent rather than viscosity of the solvent. Fluorescence lifespan of Nile blue is shortened in polar solvents and at high temperatures. Though potentially Nile red and Nile blue can both be used for the purpose of this project, Nile red would be considerably more suitable due to its neutrality allowing easy storage and its longer fluorescence lifespan, which allows a higher probability of being detected.

Traditionally Nile red has been used to stain the lipids in animal cells and microorganisms³⁶. Its lipophilicity and inexpensive price have led to the wide usage for microplastics staining and detection. The lipophilic dye binds to the polymer chain of plastics through Van der Waals interactions. In certain plastics that contain polar components in polymer chains, the dye binds through Van der Waals interactions and dipole interactions³⁷. Nile red has been reputedly known to effectively stain microplastic samples that are of types: polyethylene (PE), polypropylene (PP), expanded polystyrene (EPS), high-density Polyethylene (HDPE), polycarbonate (PC), polyurethane (PUR), and polyethylene vinyl acetate (PEVA). Shim *et al.*³⁸ referred to some of the current limitations include the weak green fluorescence stains regarding polar polymer chains such as poly vinyl chloride (PVC) and the co-staining of organic hydrophobic natural material such as chitin, seaweed, wood, feathers³⁷. These organic and hydrophobic materials showed weak fluorescence that could

pose as potential false positives during the staining process. Current suggestions were to pre-treat the samples with hydrogen peroxide or potassium hydroxide³⁹ to initiate oxidation processes. This would lead to the decomposition of natural organic materials. However, these methods were found⁴⁰ to only be somewhat effective and the oxidation process behaved similar to a bleaching effect rather than decomposition. Alternative methods such as the use of specific enzymes, cellulase, protease, chitinase⁴⁰ were used to degrade these materials and minimize false positives.

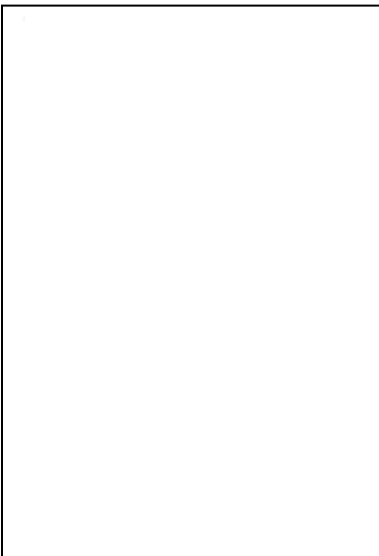


Fig. 1 The chemical structure of Nile red, drawn using Chemdraw.



Fig 2. Nile Red (NR1) with 2-propionic acid as the substituents highlighted in blue, increase in polarity for PVC binding.

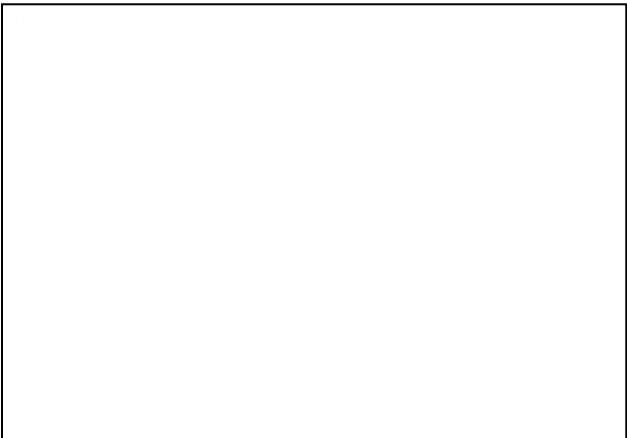


Fig 3. Nile red derivative (NR2) longer unbranched n-hexyl chain for increased lipophilicity.



Fig 4. Nile red (NR3) branched ethyl hexyl group, higher lipophilicity

An interesting article by Sturm *et al.*⁴¹ aimed to address the problem of weak fluorescence stains of polar plastic polymers, particularly PVC. The article compared Nile red, Fig. 1 with its chemical derivatives. The ethyl substituents present in Nile red were replaced with a longer unbranched n-hexyl group – named NR2, Fig. 3. Another Nile red derivative was tested, NR3, Fig. 4 replaced the ethyl substituent with branched ethylhexyl group to increase the lipophilicity of Nile red. This allowed the derivatives to bind with various sorts of microplastics with higher affinity, thus a stronger fluorescence illumination. In addition to using Nile red derivatives, the research team also prepared their testing water samples and adjusted their filter sets. Results were, in a polar solvent water, PE and Copolyester, Nile red adsorbs stronger onto non-polar plastic than if the plastic samples were in organic solvents. The research team concluded that the derivatives of Nile red, NR1 and NR2 showed a very strong signal for PP. This would hinder the identification of other microplastics. It was added that a combination of NR and NR3 in acidic water (pH 2.5) would be the best option. NR, NR1, NR2 and NR3 were tested and for each Nile red dye the emission filter was changed for the fluorescence signal test for PVC. Despite such efforts, PVC remained weakly signalled under green fluorescence detection, but at green fluorescence – wood, chalk and chitin would signal intensively due to their polar structure. Orange and red fluorescence was deemed unsuited for samples containing non-polar polymers and natural particles (wood, chalk, chitin), as the non-polar polymers would have a weaker signal than natural particles present. The paper mentioned the weak fluorescence signals for wood and chalk at longer wavelengths would be excluded by the emission filter set. Thus, wood and chalk would not be the cause of false positives. Hence, to reduce false positives, pre-sample treatment would be a reasonable and a practical approach before adding fluorescent stain and analysis.

MACHINE LEARNING ALGORITHMS

Artificial intelligence (AI) is a broad umbrella term for ‘machine learning’ in which ‘deep learning’ is a subtopic of machine learning⁴². Methods and pre-built algorithms introduced below such as Edge detection and R-CNN (region-based convolution neural network), involve ‘neural networks’ which is a subtopic of ‘deep learning’. Each computing language has its main uses, for example: Java is used for designs, C++ is used for software applications, HTML is used for website creation whilst Python is used for machine learning. Python is popular for building machine learning models because of its

simplicity and extensive libraries that support more complex tasks. The purpose of this project is to build a model allowing the quick and accurate quantification of microplastics in an image captured with a smartphone. In a further extension of the AI model would include the categorical identification of the microplastics, such as differentiating between PET and LDPE plastics and the identification of false positives present in the sample. Such tasks involve localisation of an object and classification of the object within an image and hence are known as ‘object detection’ tasks.

Dataset

The dataset used in this project are primary sourced smartphone captured images of fluorescent stained samples. All images are composed of pixels, each pixel has an RGB (Red, green, blue) value that defines its colour ⁴³. The computer ‘sees’ images based on these pixel values ⁴⁴. From the previous sections, some possible objects that would be used as ‘labels’ in the samples would include microplastics of various shapes, fibre fragments, microbeads, each with distinct colour intensity, sizes and texture features. The general shapes of the objects are used by the machine learning model to categorize them according to pre-defined labels. Whereas the colour intensity of each microplastic object could potentially be used to further categorise the type of plastic polymer. Though the larger the dataset used for model, the higher the accuracy of the model. Li *et al.*⁴⁵ experimented on training the YOLO version 2 tiny model on a small sample dataset and concluded that a dataset of less than 50 in sample size should not be used for deep learning model training. Their experiment was based on the identification of playing cards, with a sample size of 296 playing cards, the accuracy scores were around 87%. In another article⁴⁶ a dataset of 52,635 images were trained using the YOLOv4 model for facemasks detection. Their experiment achieved a mAP of 71.69%. Given the extent to which both studies were different and challenging in their own aspects, the number of images needed for this project is uncertain. In each sample image, numerous microplastic particles and various unknown objects are contained. Hence it would be best to experiment with different sized dataset and to scale up the size of the dataset if needed.

The dataset of images would need to be annotated individually to create ‘label’ classifiers. The free online website ‘Roboflow’ is commonly used for such tasks. In an article by Sarnovský and Kolárik⁴⁷ Roboflow was used to annotate lung ultrasonography images for the training of a lung disease diagnosis machine learning model. Roboflow was first launched in 2020 and is widely supported by a range of investors. The application is free to use by the public, whilst available at a price for

business start-ups and enterprises. Further studies^{48,49,50} showed that Roboflow was also used, perhaps indicating that the current-state-of-the-art image annotation for image classification machine learning models is Roboflow. With the widely available resources online, the use of this application should be feasible for this project.

Edge Detection

As opposed to a complex machine learning model consisting of neural networks, the Canny edge detection is by far a simpler method incorporated into the Python library OpenCV. OpenCV is a commonly used image processing library. The Canny edge detection algorithm was derived by John F. Canny in 1983⁵¹ and it is a simple image processing technique. The theory on the Canny edge detection algorithm is split into 3 stages. The first stage is noise reduction in the given image, by application of a Gaussian filter. The intensity gradient is then found for each pixel of the image using the Sobel kernel; a formula constituting of vertical and horizontal gradients is then applied. Once gradient values are obtained, non-maximum suppression is applied to every pixel. Unwanted pixels that do not constitute the edge are removed. The result is a thin outline of the objects in an image⁴⁴. For this project, the Canny edge detection presented above is simply just an image processing technique. Given the nature of the image dataset, without the use of machine learning techniques, the accuracy to detect false positives, microplastic and synthetic polymer identification, will significantly decrease. In the article by Rashid *et al.*⁵² various techniques combined with edge detection such as foreground extraction and colour filtering were used. However, the nature of their project was to identify ill wheat leaf, hence their image samples would've been less complex and less varied than the sample images from this project. The study by Trivedi *et al.*⁵³ used a combinational method of the Canny edge detection algorithm and the Holistically Nested Edge detection algorithm⁵⁴, in medical imaging. In the sections below, edge detection techniques are used within the hidden layers of a neural network for object detection tasks.

R-CNN, Fast R-CNN and Faster R-CNN

CNN (Convolution neural networks) whilst used for image classification tasks, R-CNN known as region based convolutional neural network first proposed by Girshick *et al.*⁵⁵ is used for object detection tasks. There are 3 main stages of R-CNN: Region proposal, Feature extraction and Classifier. The process region proposal eliminates areas in an image that do not contain classified

objects. Whilst creating bounding boxes that attempt to identify classified objects based on the colour, size, texture of previous trained data. Each bounding box contained one object cropped from the original image. These cropped images of the region proposal stage are then fed to the main body of CNN for feature extraction. The feature extraction process includes the use of kernels and matrices that creates a feature map based on the most important features in the fed object image. Pooling then occurs which 'downsamples' the feature map, generating a less pixelated object image with the most important features of the object, hence the name 'Feature extraction'. The classifier stage involves an algorithm which categorizes the object into pre-defined labels. However, the initial R-CNN model had serious time issues of analysing an image on average 32.84 seconds and obtained a 66% mAP (mean average precision); other models described below have outperformed this and would be more suited for this project. Girshick⁵⁶ overcame the time limitation and proposed the Fast R-CNN model. Instead of cropping the image into potential objects, Fast R-CNN processes the entire image with the region proposal stage with the use of edge boxes. The Fast R-CNN with VGG-16 model was tested on the Microsoft COCO dataset, a set of images of common everyday objects. The Fast R-CNN achieved on average 1.72 seconds per image at 66.9% mAP. Though the speed is acceptable for the scope of this project, the accuracy value is very poor. Hence Fast R-CNN would not be a suitable model for this project. Faster R-CNN⁵⁷ was proposed in the following year, the new model is structurally similar to Fast R-CNN with the exception of the region proposal stage. The entire image is processed, and a feature map is produced. A matrix acts as a sliding window and moves across the feature map, while setting anchors at possible objects. Each anchor placed is then finely analysed and classified into object or non-object. Then the image of 'marked' objects is passed through the main CNN and functions the same as the previous models. The faster R-CNN with VGG-16 architectural structure achieved object detection at 0.11 seconds per image at 73.2%, whilst with the ResNet-101 achieved 2.24 seconds per image at 83.8% mAP. In terms of this project, faster R-CNN (ResNet-101) would be suitable for microplastics quantification and identification. Though the identification rate is on average at 2 seconds per image, the accuracy of 83.8% is of good standard. However, in terms of feasibility, the faster R-CNN model would be time-consuming and complex to build, train and test.

You Only Look Once (YOLO)

The YOLO algorithm is one of the most powerful, current state-of-art, real-time object detection algorithms. YOLO differs from traditional methods such as RNN in the way that the algorithm is simplified and processed as a single regression problem. Early versions of the YOLO algorithms had accuracy issues and struggled to detect small objects. However, the latest 2023 version, YOLOv8 enables high-speed performance without compromising on accuracy. YOLO algorithm gained much popularity due to its free open-source code, accuracy and efficiency. However, due to the wide range of the most recent YOLO versions and the involvement of various contributors, selecting the most suitable model for this project introduces an additional layer of complexity. The following paragraphs will attempt to explain in depth each version of YOLO, the structural enhancements made and their relevance to this project. However, due to the complex nature of neural networks and the involvement of numerous research teams and companies, the development of different YOLO versions, including the most recent one, has been somewhat disorganized, resulting in significant confusion. Therefore, this YOLO section of the literature survey focused on the key aspects of each version.

The first-generation algorithm (YOLOv1) published in 2016 developed by Joseph Redmon⁵⁸ from the University of Washington aimed at image analysis as a whole. The methodology was to divide an image into square boxes then to predict the probabilities from a pre-classified custom object, in terms of confidence score. This method though well-capable of achieving 45 frames per second (i.e. Able to determine classified objects in 45 images per second), struggled to identify small objects in an image, the image input resolutions were 224x224 pixels. Further, YOLOv1 struggled to accurately locate objects that were placed closely together. Hence, YOLOv1 would be unsuitable to use for the quantification of microplastics in an image due to the size of the microplastics, the closely spaced strands of fibre contaminants and the low input resolution.

Version 1 was soon replaced by YOLOv2⁵⁹, also commonly known as YOLO-9000 because this model was used over 9000 object categories and combined a large unlabelled dataset for training. Structural enhancements made included: Introducing anchor boxes, replacing the backbone framework of the model, increasing input resolution to 416x416, increasing speed, the use of batch normalisation and multi-scale training. Anchor boxes are pre-defined boxes that are adjusted differently to each image. By adjusting the size of the boxes to detect, the chances of the model detecting small objects, were higher. The backbone of the framework was replaced by Darknet-19,

a 19-layer neural network, as opposed to a handmade convolution neural network in YOLOv1. In this way, speed and stability were increased. The use of batch normalisation was to generalise the model and reduce overfitting issues. YOLOv2 was released in 2017, as to this day, more powerful YOLO versions have been released that aims at small object detection, which would be perhaps more suited for this project.

Little enhancements were made to the third version, YOLOv3⁶⁰. The 19 convolution layers of YOLOv2 were expanded to 53 layers, YOLOv3 uses a Darknet-53 backbone framework. This meant that the model became more powerful in terms of classifying images based on their features, a higher accuracy, without sacrificing much speed. Multi-scale training was also used to generalise this model, the concept was to train the YOLOv3 model with different resolution images so the model could 'view' the same objects at different sizes. Thus, enhancing the model's ability to detect smaller objects. Resolution of input images in YOLOv3 are flexible due to multi-scale training, with a max input resolution of 608x608. In previous versions, YOLOv1 and YOLOv2 used a softmax activation function, while YOLOv3 switched to a logistic function. The softmax function calculated the probability of each category, and the one with the highest probability was chosen as the prediction. For example, when the model was predicting whether a coin was 1 Euro, or 2 Euro, the softmax might have returned probabilities like 0.19 for 1 Euro, and 0.81 for 2 Euro. Since 0.81 was the highest, the model predicted the coin was 2 Euro. The logistic function would've returned a probability of near 0 or near 1 for each category, with the highest probability being chosen. Though YOLOv3 offers a 608x608 max input resolution, modern smartphone cameras are capable of capturing higher resolutions, hence YOLOv3 might not be as suitable for this project.

The fourth version of YOLO was built on top of YOLOv3 by Bochkovskiy *et al.*⁶¹ in which the structure of the neural network, the frameworks and the techniques were improved. The article includes explanations of the methods such as Bag of Freebies and Bag of Specials used to improve accuracy and the variability of the input objects in the images. In this way the YOLOv4 model could still detect objects even if distortions were present in the images. Activation function differed in YOLOv4, 'Mish'⁶² was used as opposed to Leaky ReLU⁶³ in previous versions. One can think of Mish and Leaky ReLU as terminology for a curve i.e. logistic, sigmoidal. However, since the model was training mainly based on the Microsoft Common Objects in Context (MS COCO) dataset of images⁶⁴, custom training the YOLOv4 to a specific dataset of images will pose difficulties. In the context of the sample

images used in this project, fibres, microplastic pieces, though common in scientific areas, are quite peculiar in the context of everyday common items.

The involvement of ‘Ultralytics’ increased the versatility of YOLOv5. Ultralytics introduced Pytorch as the backbone framework as opposed to Darknet. YOLOv5 was also built compatible with Google Colab. This made the installation of YOLOv5 less complex, and this model became widely accessible to the public. Since Ultralytics is a software company, the online documentation⁶⁵ has not been published in a scientific peer-reviewed journal. However, the website provides an in-depth step-by-step guide on the building a YOLOv5 model on a custom dataset. YOLOv5 upgraded its anchor-box method into an auto-anchor generation mechanism, this automatically adjust the sizes of the anchor boxes optimised for the dataset being trained. MixUp⁶⁶ is a data augmentation technique introduced in YOLOv5. The concept is to train the model on virtual examples constructed under using linear interpolation of 2 sets of sample images, in this way the model is further generalised. CutMix⁶⁷ was introduced to create a concrete comprehensive structural enhancement to the model. The idea was to take a rectangular patch in a sample image and replace the patch with a different patch in another image. Labels of this image were then regenerated considering the new weights from the newly patched image. By ‘tricking’ the AI model in such a way, the robustness, accuracy and stability of this model was increased, so that the model was not just a set of neural network nodal calculations. Much of the key features from previous YOLO versions were maintained and added onto YOLOv5. Activation functions and data augmentation techniques such as Leaky, ReLU, mosaic data augmentation was kept from previous versions. Further studies by Zhang *et al.*⁶⁸ and by Kim *et al.*⁶⁹, both studies have made small adjustments on top of the present YOLOv5 model to suit their object detection needs. YOLOv5 is suitable to use for the purpose of this project. However, to obtain a high accuracy rate, experimentation with this model would be needed.

YOLOv6 was developed by a Chinese company ‘Meituan’⁷⁰. Most of the features of YOLOv5 were changed in YOLOv6. Instead of the CPS-Darknet Pytorch based backbone, EfficientRep was used, which is used to minimise memory usage and computational costs. The basis of this design focuses on simplifying the neural network and to remove unnecessary layers which allows more precise computational efforts. They introduced an “efficient decoupled head”, which revokes the original feature of YOLOv3 and v4, regressing bounding boxes into classifying objects. Anchor-free models were used, which removes the usage of anchor boxes and led to even faster inference times. Anchor-free models are much easier to train and generalise as well. They also enhanced the “non-maximum

suppression” function for better efficiency. Loss functions were used to calculate the model’s performance levels. Though YOLOv6 was developed in 2022, YOLOv5 maintained its popularity. The extensive community support platforms, executions and tutorials are more widely available than YOLOv6. Due to this perhaps the use of YOLOv5 would be a more suitable approach for this project. In a similar fashion, YOLOv7 though with structural advancements⁷¹ and improvements on the previous models were made, did not gain much popularity as YOLOv5.

YOLOv8 was most recently developed YOLO model by Ultralytics⁷². Some key structural enhancements made from the previous versions include: Feature pyramid network, anchor free detection, path aggregation network, and soft non-maximum suppression. These enhancements are highly relevant to this project. Feature pyramid network constructs feature maps on different scales. In doing so, the model visions both the small and large objects present in an image. This is useful for this project, because a sample image would include larger fragments such as microplastics yet have smaller fragments such as dust or fibres. This feature allows the model to visualise better, hence identifying possible false positives. Anchor free detection disregards the need for bounding boxes around objects. Instead of predicting an object location and having the model to ‘draw’ a box of the object, the model directly predicts the centre of an object. This is particularly useful for this project because often objects are clustered together in an image or objects are irregularly shaped. The use of anchor free detection allows the model to pinpoint irregularly shaped objects. Path aggregation network uses skip connections, this further allows the model to identify objects of various sizes within an image. Soft non-maximum suppression is a technique used to reduce duplicate counts when objects are overlapped in an image, increasing further accuracy.

Although instant object detection is not required for this project and earlier versions would suffice for the quantification of the microplastics, the small fibre fragments and other contaminants in the urban water runoffs would require the latest model to ensure that maximum accuracy is enforced. For this project, YOLOv5 and YOLOv8 would be the most suitable models to implement. Due to the extensive range of sources and tutorials present online, building the YOLOv5 or YOLOv8 model on a custom dataset is viable. Multiple articles^{73,74,75} have made attempts to perform the same object detection tasks using various YOLO models, in search of the most suitable model to use. Rather to limit the project on just the usage of one or two models, it is potentially worthwhile to experiment the microplastic polymer type identification, false positives detection, if permitted under time constraints. It is important to find a balance when selecting a YOLO model that can perform the

quantification and identification tasks as accurately as possible without consumption of an excessive amount of computational power.

CONCLUSIONS

The formation of microplastics through primary and secondary sources and their offset into the oceans is inevitable. In a sample of urban water runoff from various households and domestic sewage, possible contents include microbeads, LDPE, some PET fragments, synthetic polymer textiles. Though various fluorescent dyes are used to stain plastics and polymers, the most commonly used dye is Nile red. Its effective staining through Van der Waals interactions and stability have led to distinctive stains in polar environments. Though certain derivatives of Nile red are present, a combination of Nile red and NR3 could be used in acidic environments to stain microplastics more effectively. Some ideal measures can be taken during sample preparation to produce an ideal or 'cleaner' image of microplastics. Measures such as the use of the addition of cellulase, protease and chitinase to degrade potential organic false positives.

A major issue present in the use of artificial intelligence and deep learning is interpretability of the results generated by the model. The main aim of this project is to quantify microplastics. Traditional image classification tasks though simpler in structure would lack interpretability of the outcome results. For example, after an image input into a CNN model, the results might be 125 microplastic fragments present in the sample image. However, apart from manually counting the microplastic fragments, there is no way of knowing how the model generated the result 125 fragments. In the case of YOLO object detection models on the other hand, generate a bounding box around each object in the output image and shows the likelihood of that object belonging to a pre-defined label. In this way, the model interpretability can be visualised.

After the reading of various machine learning approaches for object detection tasks, this literature survey focused mainly on two models: R-CNN and YOLO. In terms of accuracy both models offer roughly 70-80% mAP on ability to classify and locate the objects within an image. However, the accuracy level changes depending on size of dataset and type of sample data images. The use of machine learning algorithms to detect and quantify microplastics have been done²⁸ and have yielded promising results. The use of Nile red for microplastics tagging have also gave positive outcomes²⁹. As our project seems to be a logical extension of the works done in the previous years, with addition of using of a black box enclosed environment, the outcome of this project appears positive.

PROJECT OUTLINE

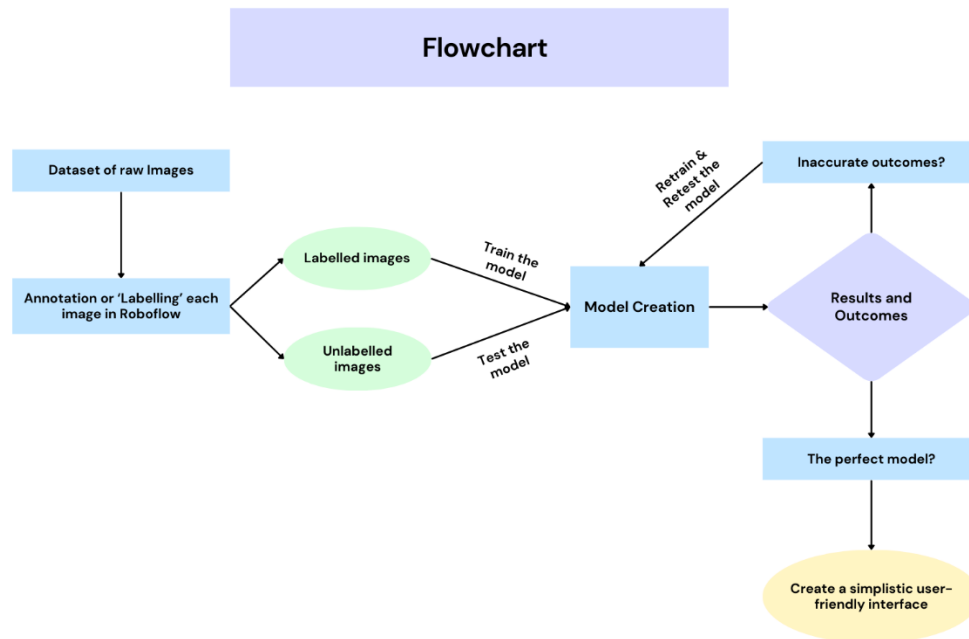


Fig. 5 Flowchart of the general process and steps to implement a model with custom trained data.

For this project, YOLOv8 will be used because it is the latest version and there's a wide range of online sources to guide the implementation of building the YOLOv8 on a custom trained dataset. Building the faster R-CNN from scratch isn't practical or feasible given the time frame of this project. One can think of a sandwich making analogy. Using YOLOv8 would be like ordering a sandwich from Subway, where toppings are pre-cut and can be added or not according to tastes. On the other hand using faster R-CNN would be like making a sandwich starting from kneading the dough of the bread, then baking it, followed by dicing toppings then assembling the sandwich.

At the beginning, roughly 50-100 sample images should suffice, preferably containing images of pure microplastics, pure fibre contaminants, mixture of both microplastics and fibres, and mixture of microplastics, fibres, natural matter such as wood, chalk, chitin and other unknown matter. The sample images used to train the model, ideally should look similar to images that the model is expected to detect. The 50-100 sample images will be then annotated using an online free open-sourced platform called "Roboflow". During the annotation stage, each item in each image will be labelled as a category. The set of 50 images will then be divided into a small portion of labelled

images used for training the YOLOv8 model. The other unlabelled images will be used to test the accuracy of the trained model. In this way, overfitting issues could be identified, and cross-validation can also be performed. If accuracy issues persist, approaches would either be that: the model would need to be trained with more sample images, the model itself needs to be enhanced, replaced with a different version or use faster R-CNN, until accuracy levels are acceptable. Once the model is somewhat created or refined, a simple user-friendly interface can be built.

Roboflow offers a simple user interface for the users to test their trained models with a individual API key. Alternatively, a wide range of libraries support user-interface builds in Python. For mobile applications, Kivy can be used, but is not recommended, as running a powerful machine learning algorithm on a mobile phone would cause loading issues or a crash in the application. An alternative and practical solution would be to build the model using Google Colab, setup Flask and Ngrok. This means that on the user side, the user opens a link on their phone browser, uploads either one or multiple sample microplastics images and outcomes are presented. Another alternative would be to use the python libraries Tkinter, PySimpleGUI or PyQt. The user would have to run the Python code on their laptop, then a popup tab will be presented with the outcomes and results laid out nicely. A fallback would be the user to have some knowledge of Python to change the file path or directory for the samples images to be processed.

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Fig. 1 The chemical structure of Nile red, drawn using Chemdraw.

Fig 2. Nile Red (NR1) with 2-propionic acid as the substituents highlighted in blue, increase in polarity for PVC binding.

Fig 3. Nile red derivative (NR2) longer unbranched n-hexyl chain for increased lipophilicity.

Fig 4. Nile red (NR3) branched ethyl hexyl group, higher lipophilicity

